

Treatment of coagulopathy in COVID-19 patients: protocol for a scoping review

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Background

Coronavirus infection disease 2019 (COVID-19) patients have a significant association with coagulopathy and disseminated intravascular coagulation contributing to high mortality. The treatment of COVID-19 and the control of coagulopathy are the primary concern. Although the evidence of anticoagulants in COVID-19 patients with coagulopathy is compiling, there is no scoping review to systematically map the anticoagulants used in this area and to identify the existing gaps between investigated treatment and not. The objective is to conduct a scoping review of available evidence on the anticoagulants in COVID-19 patients with coagulopathy.

Methods

For conducting the scoping review protocol, we will use a scoping review methodology in the Joanna Briggs Institute manual. This methodology has a five-stage approach:

(1) identify the research question; (2) identify relevant studies; (3) select studies; (4) chart the data and (5) collating, summarize and report the results. We also will use the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews to assess the results.

Ethics and dissemination

Because this is a review of the literature, ethics approval is not indicated. We will disseminate the findings from this review in publications in peer-reviewed journals.

Background

Coronavirus infection disease 2019 (COVID-19) patients have a significant association with coagulopathy. Development of coagulopathy is triggered by inflammation cytokines especially in the case of infection ([Walborn et al., 2018](#)). In addition to this basic mechanism, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infects endothelial cells. SARS-CoV-2 exploits the angiotensin-converting enzyme 2 (ACE2) receptor and induce diffuse endothelial inflammation([Varga et al., 2020](#)) resulting in thrombotic microangiopathies ([Gavriilaki and Brodsky, 2020](#)). This complex process is the characteristic coagulopathy aggravated by SARS-CoV-2. A previous observational study showed that abnormal coagulation results, especially markedly elevated D-dimer and the presence of disseminated intravascular coagulation (DIC) were frequently noted in deaths with COVID-19 patients ([Tang et al., 2020](#)).

Generally, the control of coagulopathy using anticoagulants and the treatment of underlying disease is the primary concern. Anticoagulants including unfractionated heparin, low-molecular-weight heparin or recombinant thrombomodulin have shown the effectiveness ([Murao and Yamakawa, 2019](#)) and are recommended ([Wada et al., 2013](#)).

Although the evidence of the treatment of coagulopathy in COVID-19 patients is compiling([Thachil et al., 2020](#)), there is no scoping review on this topic to our knowledge. The objective of this scoping review is to systematically map the anticoagulants used in this area and to identify the existing gaps between investigated anticoagulants and not.

Methods

In preparing this protocol manuscript, the PRISMA extension for scoping reviews (PRISMA-ScR) ([Tricco et al., 2018](#)) statement was used. This proposed scoping review will adopt the scoping review framework by The Joanna Briggs Institute (JBI) ([Peters et al., 2015](#)), which involves the following five-stage approach:

Stage 1: Identifying the research question

Stage 2; Identifying relevant studies

Stage 3; Study selection

Stage 4; Charting the data

Stage 5; Collating, summarizing and reporting the results.

Stage 1: Identifying the research question

We are interested in anticoagulants used in COVID-19 patients with DIC or coagulopathy

and the research questions are as followed:

“What anticoagulants were or will be used to treat DIC or coagulopathy in COVID-19 patients”

Stage 2: Identifying relevant studies

We used the Population, Concept, and Context (PCC) framework by the JBI for scoping review ([Peters et al., 2015](#)) to define the inclusion criteria as follows:

Patients

COVID-19 patients with coagulopathy assessed by International Society on Thrombosis and Haemostasis (ISTH) DIC score, Japanese Association for Acute Medicine (JAAM) DIC score, Japanese Ministry of Health and Welfare (JMHW) DIC score, Japanese Society on Thrombosis and Haemostasis (JSTH) DIC score, sepsis-induced coagulopathy (SIC) score even though they did not fulfill the DIC criteria or laboratory coagulation test. They receive any anticoagulants. Studies that did not report the prognosis will be excluded. Patients who received anticoagulants after having venous thromboembolism or other diseases, had coagulopathy after recovering from COVID-19 or have pregnancy will be excluded.

Concept

We will review the existing literature on the treatment and prognosis of coagulopathy in COVID-19 patients.

Context

We will not limit the context to cultural factors such as geographic location, specific racial or gender or particular settings such as acute care, primary health care or the community.

A systematic search will perform using a three-step literature search. The first step will include an initial limited search of the MEDLINE. Then, we will carry out analyses of the text contained in the titles and abstracts and the index terms of retrieved papers. In the second step to identify as many relevant studies as possible, the following databases will be searched: MEDLINE, CENTRAL, ClinicalTrials.gov, WHO International Clinical Trials Registry Platform (WHO-ICTRP) and PROSPERO. The search strategies for those databases were drafted by HM and HO. Third, we will check the reference lists of the included studies to search for other resources. We will include any study design and ongoing study registered until 21 May 2020.

Table 1. Search strategy

MEDLINE

- # 1 “disseminated intravascular coagulation”[MeSH Terms]
- # 2 “disseminated intravascular coagulation”[Title/Abstract]
- # 3 “disseminated intravascular coagulations”[Title/Abstract]
- # 4 “coagulopathy”[Title/Abstract]
- # 5 “coagulopathy”[Title/Abstract]
- # 6 “Anticoagulants”[MeSH Terms]
- # 7 “anticoagulants”[Title/Abstract]
- # 8 “anticoagulant”[Title/Abstract]
- # 9 “anticoagulation”[Title/Abstract]
- # 10 “anticoagulations”[Title/Abstract]
- # 11 #1 - #10/OR
- # 12 “Coronavirus”[MeSH Terms]
- # 13 “coronavirus”[Title/Abstract]
- # 14 “Coronavirus Infections”[MeSH Terms]
- # 15 “COVID”[Title/Abstract]
- # 16 “cov 2” [Title/Abstract]
- # 17 #7 - #11/OR
- # 18 #10 AND #17

Cochrane CENTRAL

- # 1 MeSH descriptor [disseminated intravascular coagulation] explode all trees
- # 2 "disseminated intravascular coagulation":ti,ab
- # 3 "disseminated intravascular coagulations":ti,ab
- # 4 coagulopathy:ti,ab
- # 5 coagulopathy:ti,ab
- # 6 MeSH descriptor [Anticoagulants] explode all trees
- # 7 anticoagulants:ti,ab
- # 8 anticoagulant:ti,ab
- # 9 anticoagulations:ti,ab
- # 10 anticoagulation:ti,ab
- # 11 #1 - #10/OR
- # 12 MeSH descriptor [Coronavirus] explode all trees
- # 13 coronavirus:ti,ab
- # 14 MeSH descriptor [Coronavirus Infections] explode all trees

15 COVID:ti,ab
16 "cov 2":ti,ab
17 #12 - #16
18 #11 AND #17

ClinicalTrial.gov

COVID-19

other terms: coagulation OR DIC OR coagulopathy OR anticoagulants OR anticoagulation

PROSPERO

#1 We use the following search strategy provided by PROSPERO; (((coronavirus or corona-virus) AND (wuhan or beijing or shanghai or Italy or South-Korea or korea or China or Chinese or 2019-nCoV or nCoV or COVID-19 or COVID-19 or SARS-CoV* or SARSCov2 or ncov)) OR (pneumonia AND Wuhan) or "COVID-19" or "2019-nCoV" or "SARS-CoV" or SARSCOV2 or 2019-nCov or "2019 coronavirus" or "2019 corona virus" or COVID-19 or ncov OR "novel corona virus" or "new corona virus" or "nouveau corona virus" or "2019 corona virus" OR "novel coronavirus" or "new coronavirus" or "nouveau coronavirus" or "2019 coronavirus" OR (MeSH DESCRIPTOR Coronavirus Infections EXPLODE ALL TREES) OR ((MeSH DESCRIPTOR Coronavirus EXPLODE ALL TREES) AND (wuhan or beijing or shanghai or Italy or South-Korea or korea or China or Chinese or 2019-nCoV or nCoV or COVID-19 or COVID-19 or SARS-CoV* or SARSCov2 or ncov)))

#2 disseminated intravascular coagulation" or "coagulation" or "DIC" or "coagulopathy" or "coagulopathy" or "anticoagulant"

3 #1 AND #3

WHO-ICTRP

We used the WHO-ICTRP providing all COVID-19 trials from the ICTRP database and the following terms will be used to do screening:

coagulation OR coagulations OR coagulopathy OT coagulopathy OR coagulants OR anticoagulation

Stage 3: Study selection

In the study selection process, we will use the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) flow diagram ([Moher et al., 2009](#)). We will apply the inclusion and exclusion criteria described above for the selection of studies. The process will be carried out independently by two researchers (HM and HO). When there is any disagreement, we reach an agreement by discussion. Reasons for excluded studies will be reported. We will conduct the pilot testing using a random sample of 50 citations for title and abstracts screening. If the level of agreement does not reach 75% or greater, the eligibility criteria will be modified. Modifications will be reported in detail in the full scoping review report.

Stage 4: Charting the data

We will chart the data of study designs (case report, case series, case-controlled, retrospective or prospective cohort, or RCT), coagulopathy assessed by DIC scoring systems such as ISTH DIC score, JAAM, JMW DIC score, JSTH DIC score, SIC score or laboratory coagulation test such as D-dimer or others, severity assessed by any scoring system or need for intensive care unit, anticoagulants such as prophylactic dose of unfractionated heparin, therapeutic dose of unfractionated heparin, prophylactic dose of low molecular weight heparin, therapeutic dose of low molecular weight heparin, recombinant thrombomodulin or others, outcome assessed by mortality, DIC, venous thromboembolism, multiple organ failure, coagulation laboratory data or side effects of anticoagulants. Those charting data will be mapped on the created table (Map 1). We will put included studies in order of year of publication. We will conduct the pilot testing using a random 50% sample of included studies. If the level of agreement does not reach 75% or greater, the charting data will be modified. Modifications will be reported in detail in the full scoping review report. The matrix of the map will be summarized using the following signs;

● : if the data was described and used

? : if the report did not provide sufficient information

(blank) : if the data was not used

Prognosis, in addition to study design, patients' conditions, details of anticoagulants, will be described by the study.

Map 1

				Study design					Patients' conditions						Interventions						Outcomes									
				cohort					DIC																					
				CR	CS	CC	pros.	retro.	RCTs	ISTH	JAAM	JSTH	JMHW	SIC	D-dimer	others	Severity	P-UFH	T-UFH	P-LMWH	T-LMWH	P-ENX	T-ENX	rTM	others	mortality	DIC	VTE	MOF	LCT
status	author	data	location																											
published																														
on going																														

CR: case report, CS: case series, CC: case control, pros: prospective cohort, retro: retrospective cohort, ISTH: International Society on Thrombosis and Haemostasis, JAAM: Japanese Association for acute medicine, JMHW: Japanese Ministry of Health and Welfare, SIC: sepsis-induced coagulopathy, P-UFH: prophylactic dose of unfractionated heparin, T-UFH: therapeutic dose of unfractionated heparin, P-LMWH: prophylactic dose of low molecular weight heparin, T-LMWH: therapeutic dose of low molecular weight heparin, rTM: recombinant thrombomodulin, VTE: venous thromboembolism, MOF: multiple organ failure, CLD: coagulation laboratory data

Stage 5: Collating, summarizing, and reporting the results.

First, we will present the results of the search in the PRISMA flow chart. Second, we will organize the extracted data described above. The results of this scoping review will be presented as a map, in diagrammatic form, table and descriptive format.

Differences between the protocol and the overview

Changes in the methodology that need to carry out through the study will be described in the results section.

Ethics and dissemination

This study will analyze only data of previously conducted studies, and will not involve any new studies performed by the authors. We will prepare the publication in accordance with PRISMA guideline and its adaptation for scoping reviews. We will publish our findings in peer reviewed journals.

Discussion

We believe that this scoping review will examine what researches on treatment for coagulopathy in COVID-19 is conducted and identify the gap in this topic. Systematic review is used for answering clearly defined questions, whereas scoping review is used for answering broad questions. Currently, a systematic review and meta-analysis are being conducted (Yusuf Azmi) but the available evidence is limited because COVID-19 just appeared. Review research methodologies should be used depending on the purpose of the research question (Moher et al., 2009). To our knowledge, the scoping review methodology is the best suited way to inform what researches were reported and are needed future. On the other hand, because of the urgent need for evidence on this topic, there may be more reports on anticoagulants when the results following this protocol is published.

Contribution

HM and TI conceived this study and designed the protocol. HM and HO developed the literature search strategy. HM and TI developed the charting data. HM, RT, MR and TI drafted and revised the manuscript.

Funding

No funding was obtained from any company.

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